

Day 5: Assemble and Disassemble a PC

Practical: Full Hands-on Session

Time: 1 Hour

Topic: Desktop PC Assembly and Disassembly

1. Main Components Inside a Desktop PC

Component	Short Description
Motherboard	The main circuit board of the computer. Almost all important components, such as the CPU, RAM, storage, and graphics card, are connected to it.
CPU / Processor	Often called the brain of the computer. It processes instructions and performs calculations needed to run programs.
CPU Cooler	Removes heat produced by the CPU and helps keep the processor at a safe temperature.
RAM	Random Access Memory. It is temporary memory used while programs are running and helps the computer work smoothly.
HDD / SSD	Permanent storage used for the operating system, applications, documents, photos, videos, and other files.
Graphics Card / GPU	Processes graphics and sends the video signal to the monitor. Some computers use graphics built into the CPU instead.
Power Supply Unit / PSU	Supplies electrical power to the different components inside the computer.
Cabinet / Case	The outer body that holds and protects the internal computer components.
Cooling Fans	Move air through the cabinet to help remove heat and keep components cool.
CMOS Battery	A small battery on the motherboard that helps maintain certain BIOS/UEFI settings and the system clock.
Cables	Used to provide power and transfer data between components, such as motherboard, storage, and front-panel connections.

2. Disassembling the PC

Before opening the computer, I made sure that the PC was completely shut down. I disconnected the main power cable and removed the other external cables connected to the computer.

I opened the side panel of the cabinet using a screwdriver. After opening it, I could see the motherboard, RAM, storage drive, power supply, cooling fans, and other components.

Before disconnecting anything, I checked the cable connections carefully and took clear photos. These photos can be useful later when putting the computer back together.

I carefully disconnected the cables from the different components. I removed the RAM from its slots and then removed the storage drive. If a separate graphics card was installed, I carefully removed it from the PCIe slot.

The CPU cooler can also be removed when required. After disconnecting the required cables and removing the screws, the motherboard can be carefully removed from the cabinet.

All removed screws and components were kept safely so that they would not get lost or damaged.

Photos or Diagrams

- Desktop PC before opening
- Cabinet with side panel removed
- Motherboard
- CPU and CPU cooler
- RAM
- HDD/SSD
- Graphics card
- Power supply
- Different cables
- Components after removal

3. Reassembling the PC

After completing the disassembly, I started putting the components back into the cabinet.

The motherboard was placed correctly inside the cabinet and secured with the required screws. The CPU and CPU cooler were installed correctly.

The RAM was placed back into the correct motherboard slot and pressed carefully until it was properly locked.

The HDD or SSD was placed back in its proper position and the required power and data cables were connected.

If the computer had a separate graphics card, it was placed back into the correct PCIe slot and secured properly.

After installing the main components, I connected the required power cables from the power supply. I also connected the storage cables and front-panel cables where required.

I checked the connections carefully because a loose cable or incorrectly installed component can prevent the computer from starting properly.

After checking everything, I closed the cabinet and connected the monitor, keyboard, mouse, and power cable.

Finally, I turned on the computer and checked whether it started normally.

4. Why Correct Assembly Is Important

Correct assembly is important because every component has a specific connection and position.

The motherboard needs to be securely installed so that other components can be connected properly.

RAM must be properly inserted into the slot so that the computer can detect the memory.

The storage drive needs the correct power and data connection so that the computer can access files and the operating system.

The CPU cooler must be properly installed because the CPU produces heat during operation.

Power cables must be connected correctly because different components require different power connections.

Before turning on the PC, all components and cables should be checked carefully.

5. Safety Precautions

- The computer should be completely shut down.
- The main power cable should be disconnected.
- All external cables should be removed.
- The working area should be clean and dry.
- Components should be handled carefully.
- RAM and expansion cards should be held by their edges.
- Electrical contacts should not be touched unnecessarily.

- Components should never be forced into slots.
- Screws should be kept safely in one place.
- Photos should be taken before disconnecting cables.
- The correct cables should be connected to the correct ports.
- All connections should be checked before turning on the PC.
- Liquids should be kept away from the computer.
- Anti-static precautions should be used when possible.
- The power supply unit itself should not be opened.

6. Tools Required

Tool	Use
Screwdriver	A Phillips screwdriver is commonly used to remove and install screws inside a desktop cabinet.
Anti-Static Wrist Strap	Helps reduce the risk of static electricity damaging sensitive computer components.
Small Container	Used to keep screws together so they do not get lost.
Cable Ties	Used to keep cables organized inside the cabinet.
Flashlight	Helps when looking for small screws, connectors, or cables inside the cabinet.
Clean Working Surface	Provides a safe and organized place to work with PC components.

7. Final Checklist

Before Starting

- PC is completely shut down
- Power cable is disconnected
- External cables are removed
- Work area is clean and dry
- Screwdriver is ready
- Screws can be stored safely
- Anti-static precautions are considered

During Disassembly

- Side panel opened carefully
- Internal components identified
- Photos taken before removing cables
- Power cables disconnected carefully
- Data cables disconnected

- RAM handled carefully
- Storage drive removed carefully
- GPU removed if present
- Motherboard removed if required

During Reassembly

- Motherboard installed correctly
- CPU and cooler installed
- RAM installed correctly
- HDD/SSD installed
- GPU installed if present
- Power cables connected
- Data cables connected
- Front-panel cables connected
- All screws checked
- All components checked

After Reassembly

- Cabinet closed
- Monitor connected
- Keyboard connected
- Mouse connected
- Power cable connected
- PC turned on
- Display checked
- PC starts normally

Conclusion

This practical helped me understand the internal parts of a desktop computer and how they are connected. I learned about the motherboard, CPU, RAM, storage, graphics card, power supply, cooling system, and different cables.

I also learned how to carefully handle computer components while disassembling and reassembling a PC. Taking photos before disconnecting cables can make reassembly easier. Safety is also important because computer components are sensitive to physical damage and static electricity.

After reassembling the PC, checking all connections before turning it on is important to make sure the computer works correctly. Overall, this practical gave me a better understanding of PC hardware, assembly, disassembly, and basic safety precautions.